INDEPENDENT RESEARCH / INTRADAY FORECASTING

When Does the Market Become Predictable?

Price-barrier forecasting in high-volatility equities

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A compact temporal model learns intraday price information that a time-and-volume baseline misses. The interesting part is not merely the score: it is *where* the model discriminates, how its predictions evolve, and where they fail.

5,257	0.833	0.801	18
TCN trainable parameters	-20% test ROC-AUC	-20% recent unseen ROC-AUC	Recent unseen stock-days

The result worth looking at first

I trained a small causal temporal convolutional network (TCN) to estimate remaining-session volume and the probability of large price barriers being touched. A simple multinomial logistic regression using elapsed time and cumulative dollar volume is an important sanity check: those variables alone already encode much of a session's progression. The TCN adds a compressed sequence of evolving price structure.

On the recent unseen cohort, the TCN substantially outperforms the scalar-only baseline when ranking observations *within the same retrospective volume stage*. For the Down 20% event, late-stage ROC-AUC is **0.885 versus 0.582** (60–80% stage), and **0.911 versus 0.575** (80–100% stage). These volume stages use final session volume and are therefore diagnostic groupings.

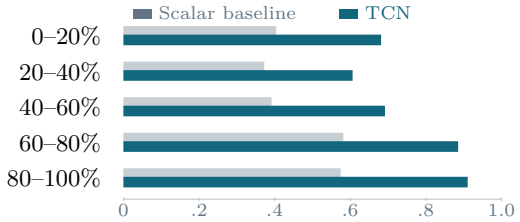


Figure 1: Recent unseen cohort: Down 20% within-stage ROC-AUC, TCN versus scalar-only logistic regression. Stages are defined retrospectively. Exact source values appear in Table 8.

What the evidence supports. The TCN learns meaningful information from evolving intraday price structure that is not captured by elapsed time and dollar-volume context alone. Its advantage over the scalar baseline is particularly pronounced when comparing observations at similar stages of a session, suggesting that the model distinguishes market states beyond simply recognizing how far trading activity has progressed.
The results also reveal an important asymmetry: downside events are generally more predictable than upside events, particularly later in the session. Performance remains strong on recent unseen observations within the model's intended market category, although probability calibration and generalization outside that category remain limitations. These findings support the model as a useful forecasting and ranking tool, rather than a complete trading strategy.

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1 A prediction in motion: IMCC

SEPTEMBER 18, 2026

Probabilities can change sharply as a session unfolds. IMCC offers a concrete example. The green lines show the model's estimated probability that the price will rise at least 20% from its current level before the session ends. The red lines show the corresponding probability of a 20% decline. Each color contains five traces, one from each independently initialized TCN. The forecasts update as the session unfolds, so both the direction of the traces and their degree of agreement are informative.



Figure 2: IMCC, September 18, 2026. The chart shows the evolving market state alongside ensemble barrier-touch probability traces.

Around 17:45 in the chart's original time display, the +20% probability traces move higher while the −20% traces move lower. The shift suggests a change in the model's assessment of the remaining session, consistent with the possibility of an upward bounce.

Each event is plotted for five independently initialized versions of the same TCN. Their separation is another observable signal: agreement can be strong in some states and much weaker in others. I use that disagreement as a practical diagnostic.

Interpretation matters. The model predicts whether specified barriers are reached before session end. It does not directly predict *when* a move will occur, whether a trade is executable, or where the optimal entry sits. These probability traces serve as a very useful indicator.

2 The forecasting problem

2.1 A narrow but demanding universe

The research focuses on high-volatility, low-float equities following intraday price expansions of roughly +70% to +500%. These sessions exhibit sharp price movements, rapidly changing liquidity, and highly uneven trading activity.

Not all expansions behave alike. I distinguish market categories by the speed and duration of the initial move, the dollar volume traded, and the price structure that develops afterward. Persistent upward moves often exhibit smooth, tightly grouped dollar-volume candles, whereas potential short-side setups tend to develop more irregular structures, with sharp reversals, unstable highs, and disrupted price action.

Experiments showed that combining these different behaviors into a broad training population weakened predictions. The model is therefore deliberately specialized for a subset of recognizable market structures rather than volatile stocks in general.

2.2 What is predicted

At each intraday observation, the model forecasts remaining traded volume and probabilities associated with future price extremes and barrier order. Four reported events are whether the remaining session reaches +10%, +20%, −10% or −20% relative to the current reference price. The barrier forecasts are derived from a multi-class extent output; an auxiliary ordering task asks about which side is touched first. Remaining-volume prediction and barrier-ordering tasks provide auxiliary supervision for the price-barrier forecasts.

2.3 What is in the dataset

Observations are generated throughout each stock-day using a fixed dollar-volume increment, rather than equal clock-time sampling. Sessions with more traded dollar volume contribute more observations. This is by design and it emphasizes states from more active days.

Table 1: Original development partitions. Stock-days do not overlap between partitions; dates do overlap because this is not a chronological split.

Partition	Stock-days	States	Covered dates
Train	368	36,176	Jan 3, 2022–Aug 25, 2026
Validation	102	10,905	Apr 5, 2022–Aug 21, 2026
Test	23	2,719	Jan 24, 2023–Aug 28, 2026

A further **23 stock-days and 4,659 states** from September 8–October 12, 2026 were evaluated without retraining and do not overlap with these partitions.

2.4 How I judge forecasts

ROC-AUC measures ranking, Brier score measures squared probability error, and ten-bin expected calibration error (ECE) summarizes calibration mismatch. Predicted and observed event frequencies help expose systematic bias. Crucially, a high AUC is not equivalent to a calibrated or actionable probability.

3 The forecasting system

A central idea behind the forecasting system is to represent market activity in **dollar space rather than clock time**. Instead of constructing candles over fixed time intervals, I use dollar-volume candles, each representing a given amount of traded dollar volume. Every candle contains OHLC prices and its duration, preserving information about both price structure and the speed at which trading activity unfolds.

This representation makes it possible to study price evolution relative to traded volume, rather than treating equal intervals of clock time as equally informative.

An important empirical finding was that how this history is compressed matters considerably. I developed an adaptive compression method that changes the dollar-volume resolution as the sequence evolves, allowing the model to retain a compact representation of the intraday price structure without letting the sequence grow indefinitely. Experiments with fixed candle resolutions and alternative representations generally performed worse.

The forecasting model itself is intentionally small: a causal, multitask TCN with **5,257 trainable parameters**. It processes the compressed candle sequence alongside elapsed time and cumulative dollar-volume context. Its

prediction heads estimate remaining-volume quantiles, future price extents, and the ordering of barrier touches. The precise adaptive compression rules and market selection criteria are omitted, while the experimental results and evaluation methodology are presented below.

4 Evidence: performance and generalization

4.1 Original validation and held-out test

The first question is whether the model can distinguish between market states with different future outcomes. The original test results show stronger discrimination for downside barriers, reaching a ROC-AUC of **0.8330** for a 20% decline.

Table 2: Barrier performance on validation and test sets. Pred., Obs., and ECE are percentages.

Barrier	ROC-AUC	Brier	Pred. (%)	Obs. (%)	ECE
Validation					
Up 10%	0.7054	0.2130	54.46	62.55	8.10
Up 20%	0.7292	0.2010	39.03	40.61	2.74
Down 10%	0.8061	0.1308	74.78	81.87	7.66
Down 20%	0.8160	0.1722	56.62	61.60	5.19
Test					
Up 10%	0.7030	0.2063	56.00	73.45	17.64
Up 20%	0.7513	0.2150	41.22	52.34	11.27
Down 10%	0.8204	0.1064	82.39	85.58	3.68
Down 20%	0.8330	0.1497	67.51	68.26	2.63

4.2 Where discrimination is strongest

Predictive performance varies considerably throughout a session. The original test results show particularly strong downside ranking in later volume stages.

Table 3: Original test ROC-AUC by intraday volume stage.

Stage	States	Up 10%	Up 20%	Down 10%	Down 20%
0–20%	531	0.621	0.694	0.732	0.779
20–40%	548	0.537	0.633	0.837	0.833
40–60%	543	0.638	0.619	0.941	0.877
60–80%	550	0.586	0.608	0.895	0.905
80–100%	547	0.558	0.546	0.819	0.832

Volume stages are defined retrospectively using the fraction of final session dollar volume, and are used for diagnostics.

4.3 Recent unseen data: the intended market category

To assess generalization beyond the original dataset, I evaluated the unchanged model on 18 new stock-days containing 2,987 prediction states, spanning September 8–October 12, 2026. These sessions belong to the market category for which the model was developed.

The results remain encouraging. Downside ROC-AUC reaches **0.8248** for the 10% barrier and **0.8011** for the 20% barrier.

Table 4: Recent unseen in-category barrier performance. Pred., Obs., and ECE are percentages.

Barrier	ROC-AUC	Brier	Pred. (%)	Obs. (%)	ECE
Up 10%	0.781356	0.179093	60.1994	74.9247	14.7487
Up 20%	0.781565	0.215391	45.6095	62.1694	16.6090
Down 10%	0.824848	0.114702	86.2975	81.0177	5.6022
Down 20%	0.801061	0.185372	72.8311	60.2277	12.6035

The stage-wise results are particularly interesting. Downside ranking strengthens late in the session, while upside ranking is strongest early.

Table 5: Recent unseen in-category ROC-AUC by volume stage.

Stage	States	Up 10%	Up 20%	Down 10%	Down 20%
0–20%	590	0.897785	0.845146	0.806795	0.788704
20–40%	598	0.833654	0.738976	0.685236	0.695604
40–60%	598	0.735260	0.617396	0.826241	0.772395
60–80%	600	0.647107	0.535565	0.969289	0.890552
80–100%	601	0.525966	0.484565	0.826210	0.882033

The 10% downside barrier reaches a ROC-AUC of **0.9693** in the 60–80% volume stage. Although this is a strong result, these retrospective stages should be interpreted as diagnostics rather than directly deployable trading conditions.

Probability calibration also remains imperfect: upside probabilities underestimate observed frequencies, while downside probabilities overestimate them. The model is therefore more useful for ranking market states than for interpreting its outputs as precisely calibrated probabilities.

4.4 Broader market stress test

I also evaluated the same checkpoint on the full recent cohort of **23 stock-days and 4,659 states**, deliberately including sessions outside the intended market category.

This provides a useful stress test of the model's behavior under a broader distribution. As expected, performance is weaker: the 20% downside ROC-AUC falls to **0.7496**.

Table 6: Broader recent unseen cohort: barrier performance. Pred., Obs., and ECE are percentages.

Barrier	ROC-AUC	Brier	Pred. (%)	Obs. (%)	ECE
Up 10%	0.7320	0.2208	52.55	73.51	20.96
Up 20%	0.7732	0.2381	37.83	59.43	21.60
Down 10%	0.7430	0.1706	85.87	74.39	11.88
Down 20%	0.7496	0.2414	69.98	50.83	19.20

Table 7: Broader recent unseen cohort: ROC-AUC by volume stage.

Stage	States	Up 10%	Up 20%	Down 10%	Down 20%
0–20%	922	0.828	0.837	0.699	0.682
20–40%	932	0.647	0.574	0.573	0.606
40–60%	935	0.682	0.646	0.738	0.692
60–80%	933	0.616	0.629	0.903	0.885
80–100%	937	0.526	0.616	0.896	0.911

Despite weaker overall performance, downside discrimination remains strong in the later volume stages. This suggests that useful predictive structure persists in parts of the broader population, even when the model is evaluated outside its specialization.

The central finding. The model generalizes more effectively within the market category it was designed to forecast. The broader evaluation illustrates the cost of applying a specialized predictor to heterogeneous market conditions.

Both evaluations use identical model parameters and preprocessing. The differences reflect the evaluation population, not changes to the forecasting system.

5 TCN versus the scalar baseline

The baseline is nine-class multinomial logistic regression using minutes elapsed since 04:00 ET, log cumulative dollar volume and log dollar-candle size. It receives *no price sequence*. Its scalar preprocessing is fitted on the training data and its regularization selected on validation cross-entropy. All comparisons use identical held-out states.

Table 8: Recent out-of-sample within-stage ROC-AUC. S = scalar regression; T = TCN.

Stage	-20% S	-20% T
0–20%	0.4042	**0.6816**
20–40%	0.3727	**0.6062**
40–60%	0.3918	**0.6918**
60–80%	0.5817	**0.8854**
80–100%	0.5745	**0.9106**

Across all five reported stages for the 20% barrier, the TCN achieves higher ROC-AUC than the scalar-only baseline. This is the strongest argument for keeping a structured representation of intraday prices rather than relying entirely on the obvious scalar signals.

6 What the experiments taught me

The model did not emerge fully formed. Some of the most useful results came from ideas that failed, particularly when additional complexity made the predictions worse rather than better.

Experiment	Observation and lesson
Adaptive versus fixed representation	Fixed dollar-candle resolution and unrestricted sequence growth performed worse in the reported experiments. The representation of market activity matters considerably.
Broader training universe	Training on all identified market categories did not improve performance on the specialized target category. More heterogeneous data was not automatically better.
Auxiliary objectives	Remaining-volume and barrier-ordering supervision were varied or removed. Removing them did not yield a consistent improvement; the final multitask setup was retained.
Larger models	Increasing capacity tended to overfit and often degraded performance, encouraging a compact network.
Added historical context	Separate historical sequence encoding and historical volume summaries did not establish reliable improvements; some approaches degraded predictions.
Alternative prediction targets	Fixed future-time and dollar-volume horizons, alternative barrier events, and multi-path forecasting did not consistently beat remaining-session barrier targets.
Price representation variants	Several smoothed price representations were worse than conventional OHLC features within the adaptive representation.

6.1 A useful surprise: more information was not always better

In this setting, detailed longer-run history and a larger network could look attractive in principle. In practice, neither reliably helped. The most productive approach was to represent the current session well, constrain the forecasting domain and validate each new modeling idea against that simpler system.

6.2 What multiple initializations reveal

A single probability trace tells only part of the story. I trained five instances of the same TCN architecture, changing only the random initialization. Although each model learns from the same data, their probability forecasts can evolve quite differently throughout a session.

The most interesting behavior occurs when these traces *diverge and converge*. Two market states may have nearly identical average predicted probabilities, yet very different levels of agreement across the five models. The spread between predictions therefore provides information that a single probability line cannot capture.

This also offers a way to investigate two sources of uncertainty. *Aleatoric uncertainty* reflects the inherent unpredictability of future market outcomes, while *epistemic uncertainty* reflects uncertainty about the learned model itself. Disagreement between independently initialized networks provides a practical, though imperfect, proxy for the latter. It does not directly measure aleatoric uncertainty or cleanly separate the two.

The evolving agreement between models is therefore interesting in its own right: it may reveal when a forecast is robust across learned solutions and when it depends heavily on initialization. These patterns improve forecasting reliability and trading decisions as we extract more information from the models.

7 Additional market examples

7.1 GCDT: when the model does not support an entry



Figure 3: GCDT, September 8, 2026. Predictions during an evolving high-volatility session.

In GCDT, the initial expansion and subsequent decline in activity could have motivated considering a short. The forecast did not provide the type of separation I was looking for, and the case is useful because it argues against acting on an appealing visual setup. It illustrates the use of the forecasts as an additional filter, not as a standalone trigger.

7.2 GRML: a stock outside the intended domain



Figure 4: GRML, September 21, 2026. A failure case with a further substantial upside move.

The model produced a misleading early configuration before the stock moved significantly higher. This session resembles a structure outside the model's intended training category.

8 What the system does not solve

8.1 Ranking is not calibration

The most encouraging performance is in discrimination: the ordering of higher- and lower-risk states. On recent unseen observations, calibration is considerably weaker. For example, the Down 20% event has a predicted frequency of 69.98% versus an observed frequency of 50.83%. Those numbers are too far apart to treat the probabilities as literal trade odds without further work.

8.2 Retrospective stage labels are not operational signals

The very strong late-stage AUC values depend on the eventual total session dollar volume, which is not known when the prediction is made. They show where information appeared in hindsight, not that the model can reliably recognize those stages online. Remaining-volume forecasts solve this problem partially.

8.3 Predicting direction is not timing an entry

Barrier-touch forecasts answer whether a move may happen before the session ends, not precisely when it begins. Entries that are early can be costly, especially for volatile short positions. Borrow availability and locate costs matter considerably.

9 Closing perspective

This research began with a simple question: can the evolving price structure of highly volatile stocks tell us something about what happens next? Answering it required more than training a neural network. Much of the work went into understanding which market behaviors were worth modeling, how to represent price action in dollar space, and which forecasting objectives could be learned reliably.

The experimental process was equally important. Larger models, additional historical information, and broader training populations did not consistently improve performance. What worked best was a compact TCN with 5,257 parameters, trained on a specific market category using an adaptive representation of intraday price structure. The resulting model demonstrated meaningful predictive discrimination on unseen sessions and captured information that a simpler time-and-volume baseline missed.

For my own trading, the result has been a useful additional indicator rather than a standalone strategy. Having studied the training population, the failure cases, and the behavior of the forecasts in detail, I understand the conditions under which I find its predictions most informative. Its main limitation is timing: the model estimates whether a price barrier will be reached before the session ends, but not when. A correct forecast can therefore still lead to a poor trade if the entry is premature.

Ultimately, the most valuable outcome is not just the trained network, but the understanding gained through building and testing it. The research has produced a practical forecasting tool, identified where its predictions carry information, and made clear which problems remain unsolved. In particular, translating accurate event forecasts into well-timed, executable decisions remains an important direction for further work.